BIG DATA ANALYTICS

TP03: Big Data Sampling

Data Quality and Interactive Analytics

Power BI Dashboard

Student: Ayoub Zekri

Academic Level: Master 2

Academic Year: 2026–2027

Module: Big Data Analytics

1 Introduction and Objectives

Big Data technologies enable the processing of massive datasets, while Business Intelligence tools transform data into useful information for decision-making. However, loading several gigabytes directly into Power BI may cause high memory consumption and slow interactive analysis.

This practical work builds on TP02 and focuses on three objectives:

- Extract a representative sample using Python and Pandas chunk processing.
- Inspect and clean data using Power Query.
- Build an interactive Power BI dashboard to analyze traffic patterns and key performance indicators.

2 Dataset and Sampling Methodology

2.1 Dataset Characteristics

The source file, `traffic_data_2016.csv`, is described as a traffic sensor dataset covering 2016. It contains more than 55.4 million records and occupies approximately 5.2 GB.

Processing a smaller sample makes interactive analysis more manageable on a personal computer.

Characteristic	Description
Source size	More than 5.2 GB
Original records	Approximately 55.4 million
Sample size	200,000 records
Sampling tool	Python and Pandas
BI platform	Microsoft Power BI

2.2 Sampling Strategy

The sampling script reads the source file in chunks instead of loading all records into memory simultaneously. A systematic sampling interval of $K = 677$ is used, with an offset maintained between chunks to preserve the sampling sequence.

For a chunk containing n records, selected positions are:

$$i_j = r + jK, \qquad i_j < n$$

where r is the first valid position determined by the sampling offset.

The selected records are combined and exported to `traffic_sample_200k.csv`, with a target of 200,000 rows.

3 Data Ingestion and Quality Profiling

The sampled CSV file was imported into Power BI and examined using Power Query Editor.

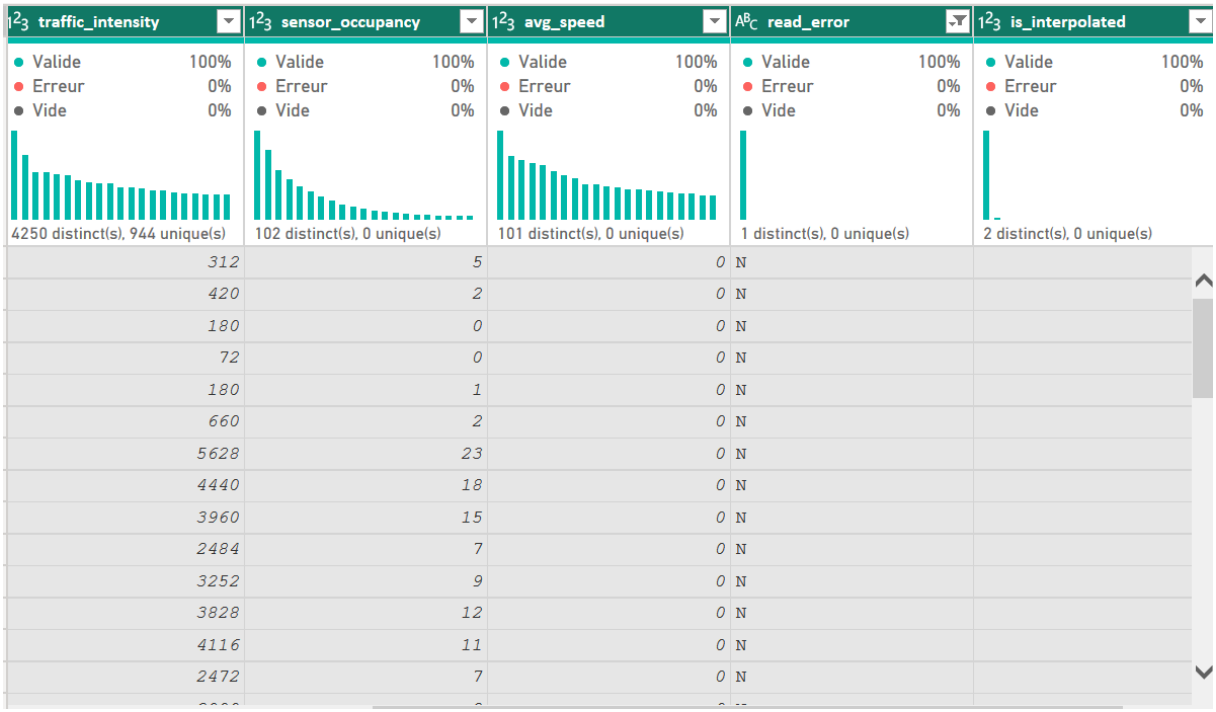


Figure 1: Power Query data profiling and column quality

The inspection focused on missing values, column validity, distinct values, and the distributions of traffic intensity, sensor occupancy, and average speed.

3.1 Data Cleaning

The sample contained 18,219 records with null `read_error` values and the reported `is_interpolated = 111` condition.

The cleaning step retained records satisfying:

$$\texttt{read_error} = \text{'N'}$$

The resulting dataset contains:

$$200,000 - 18,219 = \boxed{181,781 \text{ records}}$$

This corresponds to approximately 90.89% of the original sample. The filter is based on the stated error condition; the remaining data should not be considered fully validated without additional checks on all relevant columns.

4 Interactive Dashboard and Results

The cleaned dataset was used to create an interactive dashboard in Power BI. It combines summary indicators and temporal charts to support traffic analysis.

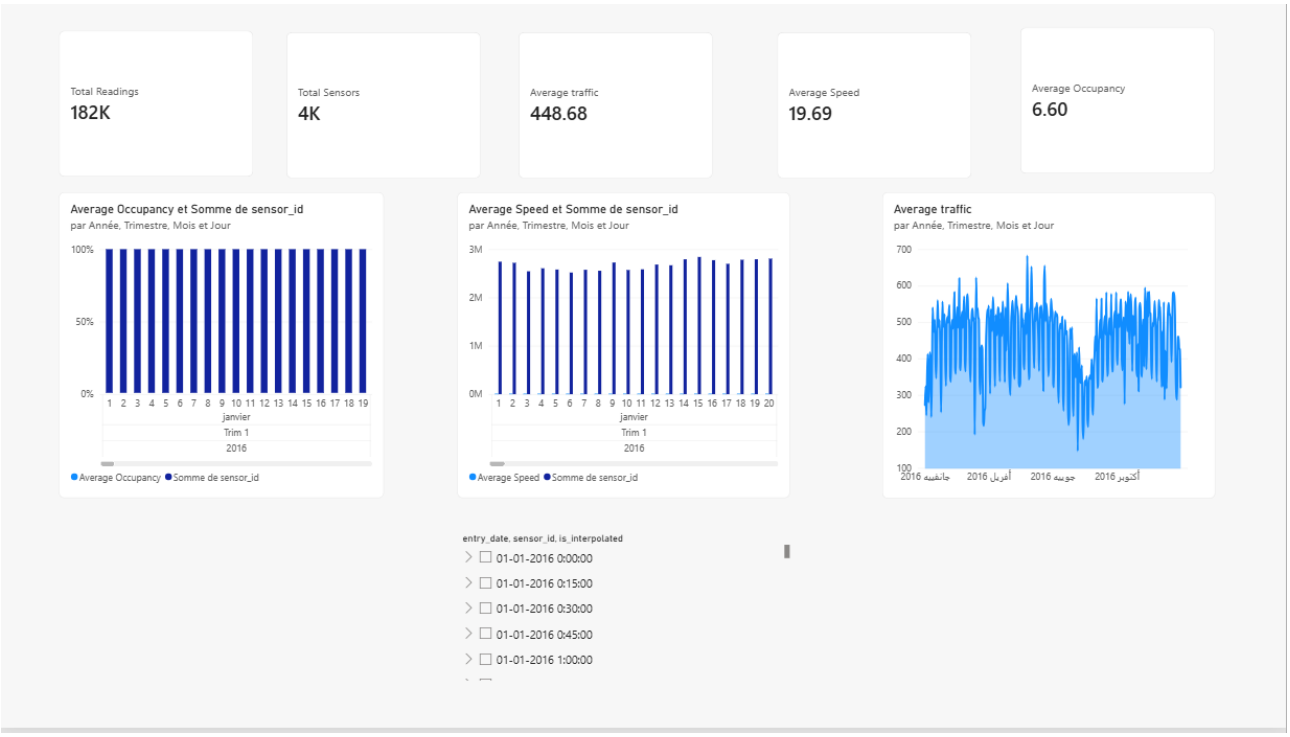


Figure 2: Interactive Power BI dashboard for traffic analytics

4.1 Key Performance Indicators

Metric	Result
Valid traffic readings	181,781
Distinct sensors	3,918
Average traffic intensity	448.68
Average speed	19.69 km/h
Average sensor occupancy	6.60%

4.2 Main Findings

- **Monthly patterns:** Average traffic intensity reached 499.23 in June and decreased to 335.34 in August, before recovering to 457.85 in September.
- **Daily patterns:** The reported minimum traffic occurred at 04:00, with an average intensity of 66.59. Higher values were observed during the day and evening, including 651.73 during the reported evening peak.
- **Weekly patterns:** Saturday and Sunday had lower reported average traffic intensities of 363.69 and 325.88, respectively.
- **Traffic indicators:** Average speed and sensor occupancy provide complementary information about traffic conditions and can be explored interactively.

These findings describe the sampled and filtered data and should not automatically be generalized to the complete traffic network.

5 Comparison Between TP02 and TP03

Aspect	TP02	TP03
Main goal	Benchmark data processing	Interactive analysis
Data scale	Full large dataset	200K-row sample
Tools	Pandas, Chunking, Dask	Python, Power Query, Power BI
Output	Performance results	Interactive dashboard

TP02 focused on processing performance and memory usage. TP03 demonstrates how sampling and data cleaning can prepare large datasets for interactive visual exploration.

6 Conclusion

This practical work demonstrates a workflow for making large traffic datasets easier to analyze using Python and Power BI.

The source data was reduced to a target sample of 200,000 records. After applying the specified cleaning filter, 181,781 records remained for dashboard analysis.

The dashboard summarizes traffic intensity, sensor count, average speed, occupancy, and temporal variations. The results illustrate how Big Data preprocessing and Business Intelligence can work together to support data exploration and reporting.